

EquiLearn: BLV Math Learning with Haptic Feedback, Audio Sonification and AI Assistance

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Abstract—Every day, students who are blind or have low vision (BLV) face barriers when it comes to education, especially in Science, Technology, Engineering, and Mathematics (STEM), where teaching comes from the standpoint of visually seeing and understanding spoken and written materials. While assistive tools and technology exist, they fail to convey enough information in real time and are often costly, specialized, or limited. Therefore, we developed EquiLearn, an interactive iOS application designed to aid BLV college students to explore and understand graph concepts independently through haptic feedback and audio sonification by leveraging widely available mobile devices, iPhones and iPads. By combining both sounds and vibrations, EquiLearn aims to reinforce classroom learning, help build confidence, and reduce reliance on accommodations in STEM. EquiChat, the application's AI-assisted explanation system, was designed to support moments of uncertainty that cannot easily be resolved through exploration alone. We designed a fully accessible math keyboard and native interface built completely around VoiceOver compatibility, enabling blind smartphone users to input complex functions touch-free. User and partner evaluation for EquiLearn was conducted iteratively across multiple sessions throughout the semester, beginning early in the discovery phase and continuing through active development. Interested readers might watch a YouTube video introducing the EquiLearn app at <https://youtu.be/wcbv7LVbQtw>.

Index Terms—Assistive Technology, Blind and Low Vision, Swift, Graph Reading.

I. INTRODUCTION

According to the World Health Organization (WHO), at least 2.2 billion people worldwide experience vision impairments, with millions of them children and adolescents [2]. Beyond visual barriers, access barriers include the lack of accommodations for students, inaccessible course materials, and the lack of knowledge by teaching professionals on how to teach blind and visually impaired students. While there are many access barriers across all levels of schooling and beyond, this paper will focus on barriers to access specifically for blind or low vision (BLV) students in higher education.

From K-12 to college, the structural and legal framework for academic accommodations changes completely. In K-12, students are supported through IEP/504 plans with more readily

available real-time support from the Teachers of the Visually Impaired (TVIs), who assist with aiding students in learning and understanding course content, and are immensely helpful in STEM courses, coupled with various learning resources. However, once students reach higher education, those plans are stripped away, and the resources available to them depend heavily on a college's funding for accessible technology and staffing. BLV students are largely left to rely on themselves and the school's limited network. These barriers to access result in low confidence from BLV students to pursue and succeed in STEM careers [1].

As a result, BLV students rely heavily on their other senses, particularly touch and sound, to learn and navigate their environment. Even while doing so, BLV individuals still experience setbacks and difficulties. Despite advances in assistive technology, existing tools for students fail to effectively convey necessary information in real time from professors to students. Current solutions range from widely accessible and affordable to highly specialized and costly, including, but not limited to, TVIs, Desmos, braille readers, screen readers, talking calculators, magnifiers, cork boards, and swell paper, which are limited in their ability in enabling blind and low-vision students to learn in real time. The lack of effective tools not only affects academic success but also the independence and confidence of visually impaired learners. As technology continues to advance and redefine how students learn, there has been a continued rise in the exploration of new channels and methods for delivering knowledge to students through non-visual methods.

To bridge this gap, our system, dubbed *EquiLearn*, provides three major contributions:

- **Dynamic Multimodal Mapping Architecture:** We developed an active mathematical rendering pipeline that translates coordinate positions into intersecting auditory pitch variations (*y*-value sonification) matched with localized proximity haptic responses.
- **Context-Aware Conversational Interface:** We integrated an AI-assisted explanation system capable of parsing mathematical equations alongside active user properties (zoom level, coordinate focus) to provide plain-language spatial descriptions.
- **Accessibility Ecosystem Integration:** We designed a

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VoiceOver-compatible math keyboard and native interface built completely around its compatibility with VoiceOver, enabling blind smartphone users to input complex functions with voice.

This paper is organized as follows: Section II presents related work in multimodal haptics and blind and low-vision (BLV) mathematical accessibility. Section III details the system design, hardware framework, and user interaction workflows of the EquiLearn application. Section IV presents an evaluation of the system, detailing qualitative feedback from field-level usability testing with expert consultants and BLV users. Finally, Section V concludes the paper and outlines prospective directions for future research.

II. RATIONALE AND RELATED WORK

A. Core Components

Graphs are inherently spatial and visual representations of information. Sighted learners are able to immediately perceive trends such as slope, curvature, intersections, and relative movement through visual pattern recognition. BLV users, however, are often required to reconstruct these same relationships sequentially through screen-reader descriptions or static text alternatives. Because mathematics instruction relies so heavily on complex visual content, traditional linear methods present severe access barriers and impose an immense, unnecessary cognitive load on BLV learners as they attempt to mentally map spatial structures sequentially [3], [8]. With EquiLearn, our goal is therefore not simply to translate graphs into nonvisual formats, but to reduce the cognitive effort required for BLV users to understand mathematical relationships. EquiLearn adopts a multimodal interaction model in which auditory, haptic, and LLM assistance each communicate different aspects of the graph. This design aligns with established research demonstrating that unimodal learning systems are consistently less effective than systems that offer multimodal approaches when providing access to nonvisual mathematical resources [3], [8].

One of the core components of EquiLearn is *EquiHertz*, for audio sonification. It was selected because continuous pitch variation naturally communicates directional change and rate of change over time. As a user moves across the graph canvas, increasing pitch corresponds to increasing y-values while decreasing pitch corresponds to decreasing y-values. This allows users to perceive the “shape” of a function dynamically rather than memorizing isolated coordinate values. As shown by Zhang et al. [7], line graphs sonification strategies employing pitch variation have been widely adopted by many other products and research efforts.

Another core component of EquiLearn is *EquiPulse*, for haptic feedback. This functionality was incorporated to provide spatial confirmation and reduce uncertainty during graph exploration. By combining haptics, users can understand their relative location to the curve and the graphing region itself. Our program addresses this by using vibration patterns to indicate curve contact and graph boundaries. By structurally

distributing the graph’s spatial parameters across both auditory and haptic pathways, the interface minimizes cognitive load and prevents a single sensory channel from becoming overwhelmed. This design choice is supported by prior research on tactile interaction for accessible graphics [4]–[6], [8]. Tennison et al. [4] investigated how blind and visually impaired users interpret specific touchscreen vibration frequencies, demonstrating that users can reliably map haptic profiles to traditional tactile line elements. Their psychophysical testing established that very low frequencies (2.5 Hz) are most consistently perceived as dashed lines, mid-low frequencies (10 Hz) as dotted lines, and mid-range frequencies (50 Hz) as constant, solid lines. EquiPulse directly uses similar concepts to deliver distinct, distinguishable tactile structural boundaries.

Because VoiceOver is the primary navigation system used by many BLV users, EquiLearn was redesigned around full VoiceOver compatibility. Direct Touch support, coordinate announcements, and axis reference cues were implemented to ensure that users could navigate graph space independently while preserving the precision needed for mathematical exploration. Existing literature on assistive mathematics technologies consistently emphasizes that accessibility tools must integrate with established assistive ecosystems rather than requiring users to adopt entirely new interaction paradigms [3].

EquiChat, the application’s AI-assisted explanation system, was developed to address moments of uncertainty that may not be resolved through exploration alone. Unlike traditional accessibility tools, which often provide static descriptions that are disconnected from the user’s current interaction state, EquiChat generates context-aware explanations using the equation, user position, quadrant, and zoom level. Users can therefore ask questions grounded in their immediate exploration, helping reduce confusion while avoiding the need to restart or reorient during graph navigation.

B. Competitive Landscape

The decision to combine these elements was motivated by both prior accessibility research and a distinct gap in existing consumer applications. This lineage of research spans over two decades; early foundational studies proved that BLV users could successfully interpret and extract information from computer-generated line graphs through touch alone. However, early tools relied on heavy, non-portable hardware [3]. While the field has steadily transitioned toward digital options, a systematic evaluation of current mobile software shows that existing smartphone applications still struggle to provide an easy navigational mechanism for continuous independent learning. This transition to interactive, digital accessibility platforms is vital, as dynamic multimodal interfaces have been shown to directly enhance student engagement, self-efficacy, and abstract concept mastery over traditional, static assistance formats [8]. To evaluate where EquiLearn sits relative to current technology, Table I shows its comparison against standard industry tools and recent research.

TABLE I
COMPARISON TO COMPETITORS

App / Tool	Platform	Audio	Haptics
Desmos [9]	Web/iOS	✓	–
SAS Graphics Accel. [10]	Web	✓	–
Talking Graphing Calc. [11]	Calculator	✓	✓
IVEO Touchpad [12]	Hardware	✓	✓
MAIDR [13]	Desktop	✓	–
AutoVizuA11y [14]	Web	–	–
EquiLearn	iOS	✓	✓

III. DESIGN AND DEVELOPMENT

A. System Engineering and Architecture Stack

EquiLearn is programmed natively in Swift and SwiftUI, targeting iOS 17+ and iPadOS 17+ environments. The processing pipeline is structurally distributed across specialized local frameworks and cloud APIs to minimize latency during live touchscreen tracing.

Local audio processing is driven by the *AVFoundation* framework, which maintains a real-time sonification pipeline mapping dynamic mathematical coordinate values to variable pitch frequencies. Localized tactile responses are powered by the *CoreHaptics* API, interfacing directly with the device’s physical Taptic Engine to provide proximity-based spatial feedback. Persistent client-side configurations and state are managed locally using the *SwiftData* engine.

Remote database infrastructure and user profile indexing are decoupled via *Firebase* and *Cloud Firestore*. Multimodal conversational math metadata handling is offloaded via an asynchronous API gateway connected to Google’s *Gemini* language model backend.

B. Application Feature Specifications

The platform provides ten core accessibility features:

- **Haptic Feedback:** Continuous proximity-based vibration whose intensity rises as the user’s touch nears the curve, with adjustable strength levels (low, mid, high), via *CoreHaptics*.
- **Audio Sonification:** Maps function y -values to real-time pitch, with low (100–400 Hz), mid (200–800 Hz), and high (400–1,200 Hz) profiles.
- **Equichat:** Parses the equation, current coordinates, and zoom level into plain-language descriptions read aloud by *VoiceOver*.
- **Interactive Onboarding Tutorial:** A four-module walk-through covering the canvas, sonification, equation input, and the assistant.
- **High Contrast Display Mode:** Renders a neon green plot line on a black canvas for low-vision users.
- **VoiceOver Integration:** Structured focus order across screens, with quadrant alerts on curve crossings and coordinate readouts on double-tap.
- **Accessible Mathematical Keyboard:** A custom symbol layout (exponents, trig functions, constants) replacing standard iOS text input.

- **Variable Equation Input:** Manual typing, voice-to-text, and quick-pick presets.
- **Persistent Canvas InfoBar:** An always-visible bar showing coordinates (x, y) , zoom level, and quadrant.
- **Centralized Accessibility Settings:** Persists haptic intensity and audio pitch preferences across sessions.

C. Graphical Interface and Interactive User flow

The baseline application user flow follows a structured sequential process across clear operational pathways (Fig. 1):

1) *Home Interface and Onboarding Sequence:* When the application icon is selected from the iOS home screen (Fig. 1a), the user lands on the primary Home Page panel (Fig. 1b). This interface presents two clear navigational entries: an interactive onboarding tutorial mode and a direct graph exploration pipeline. Selecting the interactive tutorial launches a full-screen guided experience (Fig. 1c). The onboarding pipeline breaks features down across four distinct modules to prevent cognitive overload:

- Module 1 (Explore the Canvas):* Introduces localized haptic responses on a live workspace grid (see Fig. 1d), processing boundary ticks and quadrant verbal announcements.
- Module 2 (Compare Graph Sounds):* Builds auditory literacy by allowing users to toggle between standard linear lines ($y = x$) and quadratic parabolas to note pitch changes (Fig. 1e).
- Module 3 (Enter an Equation):* Demonstrates live equation submission via quick-pick presets, allowing real-time multimodal tracing execution loops (Fig. 1f).
- Module 4 (Your Graph Assistant):* Demonstrates the conversational interface layout (Fig. 1g), displaying static chat configurations paired with an interactive AI disclaimer block.

Upon conclusion of the tutorial sequence, the interface returns the user to the home tab workspace layout, updating the dashboard to indicate a completed tutorial state with optional reset capabilities (Fig. 1h).

2) *Graph Explorer and Conversational Math Interfaces:* Transitioning to the Explore tab opens the native Graph Explorer interface (Fig. 1i). The workspace plot area allows active finger-drag rendering queries with persistent coordinate pairs visible below the grid canvas bounds. For individuals with low-vision tracking limits, High Contrast Mode transforms the panel configuration to display an optimized neon curve track (Fig. 1j). Tapping the canvas chat node reveals the context-aware Math Assistant slide-over overlay panel (Fig. 1k). This dashboard parses current canvas model rules proactively to enable free-form textual or voice queries regarding slope behavior, local minima, and vector intersections. General application preference states are tailored inside the Settings profile dashboard (Fig. 1l), exposing continuous toggles for *CoreHaptics* intensity ranges and dynamic audio frequency bounds.

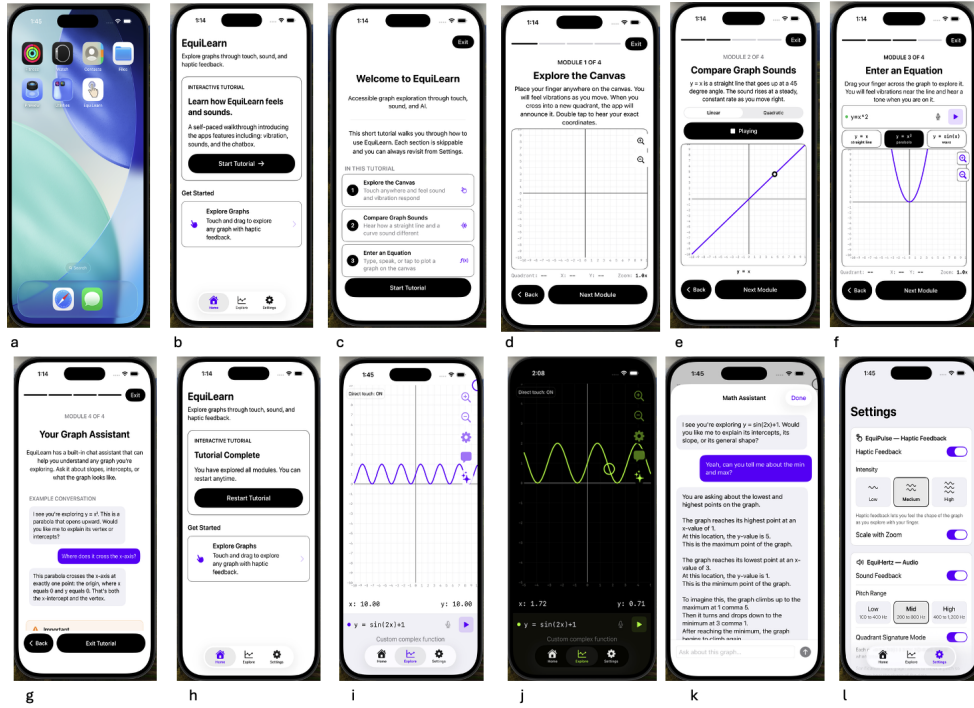


Fig. 1. EquiLearn interface. (a) Home Screen (b) Home Page (c) Interactive Tutorial (d) Module 1: Explore the Canvas (e) Module 2: Compare Graph Sounds (f) Module 3: Enter an Equation (g) Module 4: Your Graph Assistant (h) Tutorial Restart (i) Graph Explorer (j) High Contrast Mode (k) AI Assistant (l) Setting Page.

IV. RESULTS AND EVALUATION

User and partner evaluation for EquiLearn was conducted iteratively across multiple sessions throughout the semester, beginning early in the discovery phase and continuing through active development. The first round of engagement took place through the campus’s AccessAbility Center, where the team conducted initial discovery interviews to understand the tools at the disposal of students at our unique AccessAbility Center at The City College of New York (CCNY). These sessions surfaced early understandings of the inadequacy of existing accessibility features and tools for BLV students. Following the accessibility center sessions and market research, the team met with Richard Perez, a low-vision CCNY student, to gather more targeted feedback on the iOS experience specifically. Perez identified concrete constraints around how low-vision users interact with iOS systems and pinpointed areas where, based on his experience helping design accessible apps, our existing accessibility falls short for tasks such as noting font sizes, button sizes, and colors. These early meetings with Perez helped the team understand the type of thought process needed for an accessible, user-centric app.

We then met with Perez again with a more concrete user interface, and took note of feedback such as integrating a color wheel to account for various types of low vision impairments. The team then conducted an in-person accessibility validation session with Chancey Fleet, Assistive Technology Coordinator at the Andrew Heiskell Braille and Talking Book Library at the New York Public Library, alongside two additional BLV

individuals. Fleet served as an expert consultant rather than an end user. Her role was to evaluate the application and accessibility architecture of the prototype against professional AT standards. This session produced the most substantial design feedback of the semester, directly shaping the double-tap coordinate system, the tutorial structure, the AI disclaimer, and the Direct Touch VoiceOver implementation.

The team subsequently met again with Perez for a follow-up session, he suggested improvements to our high-contrast mode color scheme to provide a better view of the app, as well as adding certain font types that are easier to read for low-vision individuals.

An additional consultation was held with Hong Zhao, CCNY. As an expert advisor on blind-user interface habits, Hong pointed out a major flaw in our testing: BLV users tend to swipe across screens very rapidly, meaning they will completely miss a thin haptic line if they have to find it purely by accident. He suggested adding a point-by-point guidance mode using clock directions (e.g., telling the user to move toward “3 o’clock” to follow the curve). He also advised stripping Markdown characters from the AI text outputs so VoiceOver wouldn’t read aloud broken punctuation, and suggested recording touch coordinates during tests to see exactly where users get lost on the screen.

The evaluation cycle concluded with a final session involving both Fleet and Perez to test the core features. Based on their testing with VoiceOver active, Fleet pointed out that continuous haptic vibration quickly causes finger fatigue and

drains the battery; she recommended replacing it with brief “bumps” on the axis and edges. The team also noted that the current haptic frequency might feel jarring or creepy to some users. For the audio interface, we were advised to resolve a “choppiness” issue in the VoiceOver playback and implement a timeout rule for the voice mode so users do not have to constantly toggle it on and off.

Finally, to polish the usability, we also got feedback regarding missing button descriptions and added math signs directly to the primary keyboard for quicker input. With Perez, we also got more feedback on how we could improve our features for low vision individuals, including what color combinations would work best, and more insights on how low vision individuals might use some accessibility features differently compared to blind users. Across all evaluation sessions, the EquiLearn team used feedback not as validation but as direct input into the build cycle. Features were added, restructured, or reprioritized based on what emerged from each session, reflecting a continuous partnership between development and the communities the app is designed to serve.

V. CONCLUSION AND DISCUSSION

No single modality fully captures all aspects of graphical understanding. In EquiLearn, audio effectively communicates continuous mathematical trends, haptics provide positional certainty, VoiceOver supports structured navigation, and conversational AI offers contextual clarification. Together, these systems reduce the mental effort required to interpret graphs nonvisually while supporting multiple interaction preferences and accessibility needs. EquiLearn is free because cost is itself an accessibility barrier.

That said, our application is built to function using Apple products and targeted towards college students. While iOS provides a simple framework with great accessibility features like VoiceOver and CoreHaptics, Apple devices remain costly and are not universally accessible to all students. Extending EquiLearn to Android and to the web would remove that remaining dependency and is one of our strongest future goals.

Future work may also explore how EquiLearn behaves on larger touch surfaces. The application already runs on iPad, but the tactile channel is unavailable there because these devices lack the Taptic Engine that drives haptic feedback on iPhone. Restoring that channel through an external actuator or a paired haptic accessory would give students a larger canvas on which to trace a curve as well. Extending the same interaction model further, to smart whiteboards, monitors, and televisions, would allow an instructor and a student to explore a single graph together on a shared surface.

Beyond refining our existing features, after more pilot testing, the team is interested in exploring how we can build upon EquiLearn to provide a more holistic user experience that helps students learn beyond simple graphs. Mathematics is a multidimensional subject, and a student’s ability to comprehend, apply, and feel confident is tied to how well they can engage with concepts at multiple levels. The goal is not

just to trace a curve, but to understand what it means, how it behaves, and how it connects to broader mathematical ideas.

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REFERENCES

- [1] National Federation of the Blind, “The Braille literacy crisis: A report to the nation on the importance of literacy in the 21st century,” 2009. [Online]. Available: https://nfb.org/images/nfb/documents/pdf/braille_literacy_report_web.pdf
- [2] World Health Organization, “World report on vision,” Geneva, 2019.
- [3] M. Shoaib et al., “Assistive technology-based solutions in learning mathematics for visually impaired people,” *Multimedia Tools and Applications*, vol. 82, no. 25, pp. 41913–41950, 2023.
- [4] J. L. Tennison, P. M. Uesbeck, N. A. Giudice, A. Stefik, D. W. Smith, and J. L. Gorlewicz, “Establishing vibration-based tactile line profiles for use in multimodal graphics,” *ACM Transactions on Applied Perception*, vol. 17, no. 2, pp. 1–14, 2020, doi: 10.1145/3383457.
- [5] J. L. Toennies, J. Burgner, T. J. Withrow, and R. J. Webster, “Toward haptic/aural touchscreen display of graphical mathematics for the education of blind students,” in *2011 IEEE World Haptics Conference*, pp. 373–378, 2011.
- [6] W. Yu, R. Ramloll, and S. A. Brewster, “Haptic graphs for blind computer users,” in *Proceedings of the 1st International Workshop on Haptic Human-Computer Interaction*, pp. 41–51, 2000.
- [7] Z. Zhang, J. R. Thompson, A. Shah, M. Agrawal, A. Sarikaya, J. O. Wobbrock, E. Cutrell, and B. Lee, “ChartA1ly: Designing accessible touch experiences of visualizations with blind smartphone users,” in *Proceedings of the 26th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '24)*, pp. 1–15, 2024.
- [8] M. Y. Ruamba, Y. L. Sukestiyarno, Rochmad, and T. S. N. Asih, “The impact of visual and multimodal representations in mathematics on cognitive load and problem-solving skills,” *International Journal of Advanced and Applied Sciences*, vol. 12, no. 4, pp. 164–172, 2025.
- [9] Desmos, “Desmos accessibility,” Desmos Studio. Accessed: Jul. 20, 2026. [Online]. Available: <https://www.desmos.com/accessibility>
- [10] Perkins School for the Blind, “Getting started with SAS and SAS Graphics Accelerator.” Accessed: Jul. 20, 2026. [Online]. Available: <https://www.perkins.org/resource/getting-started-sas-and-sas-graphics-accelerator/>
- [11] Orbit Research, “Orion TI-84 Plus talking graphing calculator.” Accessed: Jul. 20, 2026. [Online]. Available: <https://www.orbitresearch.com/products/blindness-products/talking-calculators/orion-ti-84-plus-talking-graphing-calculator/>
- [12] ViewPlus Technologies, “IVEO 3 hands-on learning system.” Accessed: Jul. 20, 2026. [Online]. Available: <https://viewplus.com/product/iveo-3-hands-on-learning-system/>
- [13] J. Seo, Y. Xia, B. Lee, S. McCurry, and Y. J. Yam, et al., “MAIDR: Making statistical visualizations accessible with multimodal data representation,” in *Proc. 2024 CHI Conf. Human Factors in Computing Systems (CHI '24)*, Honolulu, HI, USA, 2024. [Online]. Available: <https://arxiv.org/abs/2403.00717>
- [14] D. Duarte, R. Costa, P. Bizarro, and C. Duarte, “AutoVizuA1ly: A tool to automate screen reader accessibility in charts,” *Computer Graphics Forum*, vol. 43, no. 3, Art. no. e15099, 2024. [Online]. Available: <https://www.semanticscholar.org/paper/e9ac71bac8211cc56eb47f2672cf349ae17d3e7c>